$\sqrt{(b-a)^2} = \sqrt{(a-b)^2}$

$\sqrt{(a-b)^2} \cdot \sqrt{(b-a)^2} = |b-a| \cdot |a-b| = (b-a) \cdot (a-b)$

$-b^2 + ab + ab - a^2 = -9$

$\frac{2}{\sqrt{6}} = \frac{2\sqrt{6}}{6} = \frac{\sqrt{6}}{3}$

$\sqrt{3}$

6) A: $\sqrt{5} + \sqrt{3}$ ve $12 = \sqrt{5} + \sqrt{3}$ için $\frac{A}{12} + \frac{B}{A} - 2 = ?$

$\frac{\sqrt{3} - \sqrt{2}}{\sqrt{5} + \sqrt{3}} \cdot \frac{\sqrt{5} + \sqrt{3}}{\sqrt{5} - \sqrt{2}} + \sqrt{2} \Rightarrow \frac{(\sqrt{5} - \sqrt{3})^2}{\sqrt{3}^2 - \sqrt{2}^2} + \frac{(\sqrt{5} + \sqrt{2})^2}{\sqrt{5}^2 - \sqrt{2}^2} - 2$

$(\sqrt{3} - \sqrt{2})$ $(\sqrt{5} + \sqrt{2})$

$\frac{5 - 2\sqrt{15} + 3 + 5 + 2\sqrt{15} + 3}{2} + 2 = 8 + 2 = 10$

ç) $x<0<4$ ... $\frac{|2x-4|-|2x-x|}{x+4}$ =?

$x<0<4 \Rightarrow 2x<0$ ve $4>0$ $\quad 2x-4<0$ ve $2x+x>0$ ...

$\frac{|2x-4|-(2x-x)}{x+4}$ = $\frac{-(2x-4)-(2x-x)}{x+4}$ → $\frac{-2x+4-2x+x}{x+4}$

$\frac{-2x+4-2x+x}{x+4}$ → $\frac{-(x+4)}{x+4}$ → -1 //

d) $\sqrt{0,64}+\sqrt{0,04}-\sqrt[3]{0,027} = \sqrt{\frac{64}{100}}+\sqrt{\frac{4}{100}}-\sqrt[3]{\frac{27}{1000}}$

$= \sqrt{(\frac{8}{10})^2}+\sqrt{(\frac{2}{10})^2}-\sqrt[3]{(\frac{3}{10})^3} = \frac{8}{10}+\frac{2}{10}-\frac{3}{10} = \frac{7}{10} = 0,7$

e) $\frac{\sqrt{75}-\sqrt{27}+\sqrt{12}}{\sqrt{100}+\sqrt{64}} = \frac{\sqrt{5^2\cdot3}-\sqrt{3^2\cdot3}+\sqrt{2^2\cdot3}}{\sqrt{6^2\cdot2}+\sqrt{4^2\cdot3}} \rightarrow \frac{5\sqrt{3}-3\sqrt{3}+2\sqrt{3}}{6\sqrt{2}+4\sqrt{3}}$

$\frac{4\sqrt{3}}{10\sqrt{3}} \rightarrow \frac{2}{5}$

f) $\sqrt{2}\cdot(\sqrt{18}+\sqrt{50}-\sqrt{32}-\sqrt{8}) = \sqrt{36}+\sqrt{100}-\sqrt{64}-\sqrt{16}$

$6+10-8+4 = 12$

g) $\frac{\sqrt[3]{16}}{\sqrt[5]{\frac{1}{2}}} = \sqrt[5]{\frac{16}{\frac{1}{2}}} = \sqrt[5]{32} = \sqrt[5]{2^5} = 2$

h) $\sqrt[5]{\sqrt[3]{\sqrt{2}}\sqrt{2}} \rightarrow 5\cdot2\cdot2\cdot3\sqrt{2} = \sqrt[120]{2}$

ı) $\sqrt[3]{2}\sqrt[3]{\sqrt[3]{4}} = \sqrt[3]{\sqrt[3]{2^{2\cdot3}\cdot4}} = \sqrt[3\cdot3]{2^6\cdot2^2} = \sqrt[18]{2^8} = \sqrt[9]{2^4} = 2^{\frac{4}{9}}$

i) $A>0$; a $\quad (\sqrt[3]{a^2})^4 \rightarrow (\sqrt[3]{a^4})^4 \rightarrow \sqrt[3]{a^3\cdot a^3} = a\sqrt[3]{a^3}$

j) $\sqrt{(-4)^2}-\sqrt[3]{(-2)^3}+\sqrt[4]{(+2)^4} = \sqrt[3]{8^1} \rightarrow |-4|-(-2)+|-2|+3 =$

$= 4+2+2+3 = 11$

6/ $\left(\left(-\frac{1}{2}\right)^{-1}\right)^4 \rightarrow -\left(-\frac{1}{2}\right)^{-4} = -(-2)^4 = -2^4 = -16$

8/ $\frac{5^{30}+5^{25}+5^{20}}{5^{25}\cdot 5^{20}+5^{45}} = x \Rightarrow \frac{x}{125}$

↓

9/ $\frac{5^{20}(1+5^5+5^{10})}{5^{25}(5^{10}+5^5+1)}$ $\frac{5^{20}}{5^{25}} \rightarrow 5^{-5}$ $\frac{5^5}{5^7} \rightarrow 5^{-2}$

10/ $\frac{14^n+6^n+2^n}{7^n+3^n+1} \rightarrow \frac{2^n(7^n+3^n+1)}{7^n+3^n+1} = 2^n$

11/ $(0,027)^{\frac{5}{3}} \cdot 10^5 \rightarrow (27\cdot 10^{-3})^{\frac{5}{3}} \cdot 10^5 = (3^3)^{\frac{5}{3}} \cdot (10^{-3})^{\frac{5}{3}} \cdot 10^5$

$= 3^5 \cdot 10^{-5} \cdot 10^5 = 3^5 \cdot 10^0 = 3^5$

12/ $\frac{-9^{-2}\cdot 25^{-1}}{125^{-1}\cdot 27^{-1}} \rightarrow \frac{\cdot (3^2)^{-2}\cdot (5^2)^{-1}}{(5^3)^{-1}\cdot (3^3)^{-1}} = \frac{\cdot 3^{-4}\cdot 5^{-2}}{5^{-3}\cdot 3^{-3}} = -\frac{5^{3-2}}{3^{4-3}} = -\frac{5}{3}$

13/ $\left(\frac{a^x}{a^4}\right)^{x-4} \cdot \left(\frac{a^4}{a^x}\right)^{x-4} \rightarrow 1$

14/ $3^x = 2$ ise $27^{x+1} \cdot 81^x = ?$

$(3^3)^{x+1} \cdot (3^4)^x = 3^{3x+3} \cdot 3^{4x} \rightarrow (3^x)^3 \cdot 3^3 \cdot (3^x)^4$

$\downarrow 2$ $\downarrow 2$

$2^3 \cdot 3^3 \cdot 2^4 \rightarrow 200$

15/ $x-2=4$ ise $|x-4|+|y-x|=?$

$x-4=2$ $|x-4|+|4-x| = |2|+|-2| = 2+2=4$

16/ $-2<x<4$ ise $|x+2|+|x-4| = ?$

$-2<x<4 \rightarrow x+2>0$ ve $x-4<0$

$|x+2|+|x-4| = x+2-(x-4)$ $x+2-x+4 = 6$